Prinsip Apit

Misal $f(x) \leq g(x) \leq h(x)$ untuk x disekitar c dan

$$\lim_{x \to c} f(x) = L \quad \text{serta} \quad \lim_{x \to c} h(x) = L$$

maka

$$\lim_{x \to c} g(x) = L$$

Contoh Hitung $\lim_{x \to 1} (x-1)^2 \sin \frac{1}{x-1}$

Karena $-1 \leq \sin(\frac{1}{x-1}) \leq 1 \Rightarrow -(x-1)^2 \leq (x-1)^2 \sin \frac{1}{x-1} \leq (x-1)^2$

dan $\lim_{x \to 1} -(x-1)^2 = 0$, $\lim_{x \to 1} (x-1)^2 = 0$

maka $\lim_{x \to 1} (x-1)^2 \sin \frac{1}{x-1} = 0$

Tentukan nilai $\lim_{x \to 0} x \sin\left(\frac{1}{x}\right)$

Jawab :

$$-1 \le \sin\left(\frac{1}{x}\right) \le 1$$

$$-x \le x \sin\left(\frac{1}{x}\right) \le x$$

$$\lim_{x \to 0} -x \le \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) \le \lim_{x \to 0} x$$

$$0 \qquad = \qquad 0$$

$$\rightarrow \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$$

Limit Fungsi Trigonometri

$$1.\ \lim_{x \to 0} \frac{\sin\ x}{x} = 1$$

$$2.\ \lim_{x \to 0} \cos\ x = 1$$

$$3.\ \lim_{x \to 0} \frac{\tan\ x}{x} = 1$$

Contoh:

$$1.\lim_{x \to 0} \frac{3x + \sin\ 4x}{5x - \tan\ 2x} = \lim_{x \to 0} \frac{3 + \frac{\sin\ 4x}{4x}.4}{5 - \frac{\tan\ 2x}{2x}.2} = \frac{3 + \lim_{x \to 0} \frac{\sin\ 4x}{4x}.4}{5 - \lim_{x \to 0} \frac{\tan\ 2x}{2x}.2}$$

$$= \frac{3 + \lim_{4x \to 0} \frac{\sin\ 4x}{4x}.4}{5 - \lim_{2x \to 0} \frac{\tan\ 2x}{2x}.2} = \frac{7}{3}$$

x → 0 ekivalen dgn 4x → 0

2. Tentukan nilai $\lim_{x \to -3} \frac{1-\cos\,(x+3)}{x^2+6x+9}$

Jawab :

$$\lim_{x \to -3} \frac{1-\cos\,(x+3)}{x^2+6x+9} = \lim_{x \to -3} \frac{1-\cos\,(x+3)}{x^2+6x+9} \frac{1+\cos\,(x+3)}{1+\cos\,(x+3)}$$

$$= \lim_{x \to -3} \frac{\sin^2\,(x+3)}{(x+3)(x+3)(1+\cos\,(x+3))}$$

$$= \lim_{x \to -3} \frac{\sin\,(x+3)}{(x+3)} \lim_{x \to -3} \frac{\sin\,(x+3)}{(x+3)} \lim_{x \to -3} \frac{1}{1+\cos\,(x+3)}$$

$$= 1 \quad 1 \quad \tfrac{1}{2} = \tfrac{1}{2}$$

3. $$\lim_{x\to 0} \frac{\sin x}{\sqrt{1-x}-1} = \lim_{x\to 0} \frac{\sin x}{\sqrt{1-x}-1} \frac{\sqrt{1-x}+1}{\sqrt{1-x}+1}$$

$$= \lim_{x\to 0} \frac{\sin x\,(\sqrt{1-x}+1)}{-x} = -2$$

4. $$\lim_{x\to 2} \frac{\tan(\sqrt{x+2}-2)}{x-2} = \lim_{x\to 2} \frac{\tan(\sqrt{x+2}-2)}{x-2} \frac{(\sqrt{x+2}-2)}{\sqrt{x+2}-2} \frac{\sqrt{x+2}+2}{\sqrt{x+2}+2}$$

$$= \lim_{x\to 2} \frac{x+2-4}{(x-2)(\sqrt{x+2}+2)}$$

$$= \lim_{x\to 2} \frac{x-2}{(x-2)(\sqrt{x+2}+2)} = \frac{1}{4}$$

Soal Latihan

Hitung

1. $\lim_{t \to 0} \frac{\tan^2 3t}{2t}$

2. $\lim_{t \to 0} \frac{\cot \pi t \sin t}{2 \sec t}$

3. $\lim_{t \to 0} \frac{\cos^2 t}{1 + \sin t}$

4. $\lim_{t \to 0} \frac{\sin 3t + 4t}{t \sec t}$

5. $\lim_{x \to 0} \frac{\tan x}{\sin 2x}$

6. $\lim_{x \to 2} \frac{\sin\left(\sqrt{x+2} - 2\right)}{x-2}$

7. $\lim_{x \to \pi} \frac{\pi - x}{\sin x}$

8. $\lim_{x \to 0} \frac{x^2}{1 - \cos x}$